

ADVAITH RENJITH

Rutgers University | Mechanical & Aerospace Engineering | Class of 2030

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Research interests: fluid dynamics and physics-informed modeling; autonomous systems; soft robotics.

EDUCATION

Rutgers University, School of Engineering | B.S. in MAE, expected May 2030

Edison Academy Magnet School | Class of 2026

SAT 1580 (Math 800); AP Physics C Mechanics / E&M: 5 / 5; AP Calculus BC: 5.

RESEARCH & ENGINEERING

QuGTrans: Physics-Informed Flood Forecasting | 2025-2026

2nd Place, NJ State Science Fair - Computer Science Division

- Achieved 117× faster inference than a finite-difference reference in project benchmarks, with 97% mass-conservation adherence.
- Added the 1D Saint-Venant continuity equation to the training loss to constrain predicted flood dynamics by conservation of mass, combining fast learned forecasts with a physical consistency objective.

Soft Robotics Arm | Independent project, 2024-2026

- Built a three-cable, segmented thermoplastic polyurethane (TPU) continuum arm; used differential cable tension and material compliance for spatial actuation.
- Integrated vision-language guidance and heuristic control with pose-driven actuation for 3D trajectory tracking.

Active Aerodynamics Testbed | Independent project, 2023-2026

- Built a 1/8-scale RC platform with a ground-effect downforce fan and drag-reduction system for active-aerodynamics experiments.
- Repurposed a smartphone as the vehicle controller and telemetry unit for on-device sensing and test-data collection.

NASA App Development Challenge - Team Luna Vista | 2025

National Winner, 200+ teams

- Modeled the Artemis II flight path in Unreal Engine 5 using orbital mechanics and C++ shaders; optimized antenna scheduling for the winning visualization.

PUBLICATION

Beyond the Haystack: Sensitivity to Context in Legal Reference Recall

Third author | NLLP Workshop at EMNLP 2025 | Accepted 2025

- Designed an evaluation framework for long-context recall across three naturalistic legal-reference tasks using U.S. court-opinion briefings; exposed retrieval behavior missed by synthetic benchmarks.

TECHNICAL SKILLS

Programming: Python, C++ | **Modeling:** physics-informed learning, model evaluation, reinforcement learning

Hardware: TPU prototyping, cable actuation, RC telemetry, CubeSat design

Tools: Unreal Engine 5, CAD, on-device inference | **Safety training:** OSHA-10

HONORS & SERVICE

Sir Isaac Newton Physics Exam: 9th globally among 3,800+ participants.

USAPhO: Distinction; top 400 nationally; F=MA qualifier.

MIT Beaver Works Summer Institute: 4th nationally; wildfire-detection CubeSat design.

NJ Aviation Hall of Fame (2024-2026): digitized 8,000+ pages of aerospace archives.

InfoAge Science & History Museums (2024-2026): artifact database spanning 20 buildings; technical repairs and public interpretation.